



Worms at Work: *Turning Campus Food Waste into Climate Solutions at NAU Through Vermicomposting Systems*



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INTRODUCTION



- The Vermicompost Living Lab (VLL) was established at the Environmental Engineering Field Station on NAU's south campus to process organic waste from campus dining halls.
- The project was designed as a test model that could be up-scaled for advancing sustainable waste management practices that reduce campus waste and lower greenhouse gas emissions.
- Vermicomposting offers a biologically efficient, low energy method for advancing existing composting practices, and diverting food waste from landfills.

METHODS

- A 6 ft x 10 ft vermicomposting pit was constructed and 10,000 Red Wiggler (*Eisenia fetida*) worms were added along with cardboard and post-consumer food collected from The DüB Dining District.
- The initial post-consumer food, a byproduct of the Somat machine, contained meat, dairy, oils and led to maggots.
- Switched to pre-consumer food - added weekly and supplemented with home food scraps.



METHODS

- Compost and castings samples were collected weekly.
- Will be analyzed using various sequencing techniques (i.e., 16S amplicon, metagenomics, metatranscriptomics) to identify key enzymes, genes, and taxa to better understand composting efficiency.
- Temperatures and pH balances were collected and monitored weekly, making adjustments as needed.



- The worms were "winterized" December-February to compensate for the colder Flagstaff weather.
- Data collected was entered into a metadata spreadsheet and will be used for future analysis.

RESULTS



- A comprehensive feasibility study was conducted to outline a scalable vermicomposting plan for the NAU campus.
- Post-consumer food waste attracted pests, including maggots, highlighting the need for a separate long-term collection method or a shift to pre-consumer food waste.
- Through collaboration with Chef Bill from the DüB, collection of pre-consumer food was made easy and the worms adjusted to the change positively.



- The VLL provides opportunities for on-campus research, collaborations, and workshops.
 - SSLUG Garden - community education and outreach events, including a vermicomposting workshop to install bins at the garden.
 - Earth Jam and Earth Day events raised awareness among the broader student population about the importance of food waste sustainability.
- Public interest surveys conducted on campus and in the broader community showed favorable attitudes toward vermicompost and recognition of the economic value of castings.

CONCLUSION

- This system is scalable!
- There is public, student, and employee interest in implementing vermicomposting systems on campus.
- The dining halls produce consistent ample pre-consumer food scraps that can be used as food stock.



- This project and future expansion opens the door for students to gain hands-on research experience.
- Invites cross collaboration between departments to explore sustainable ways of waste disposal.

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